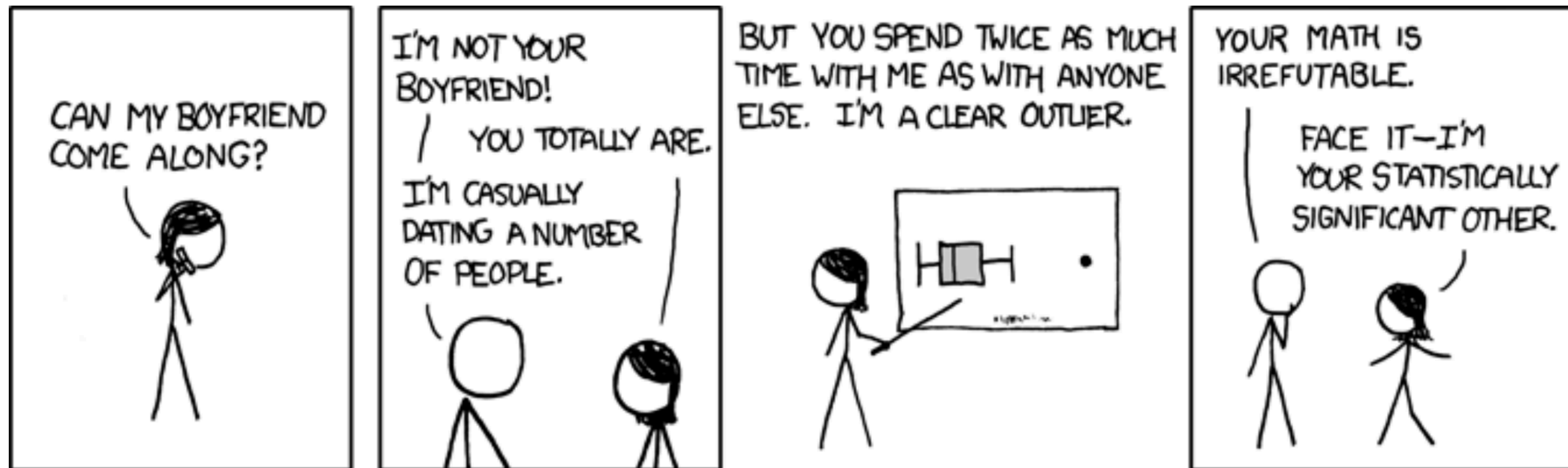
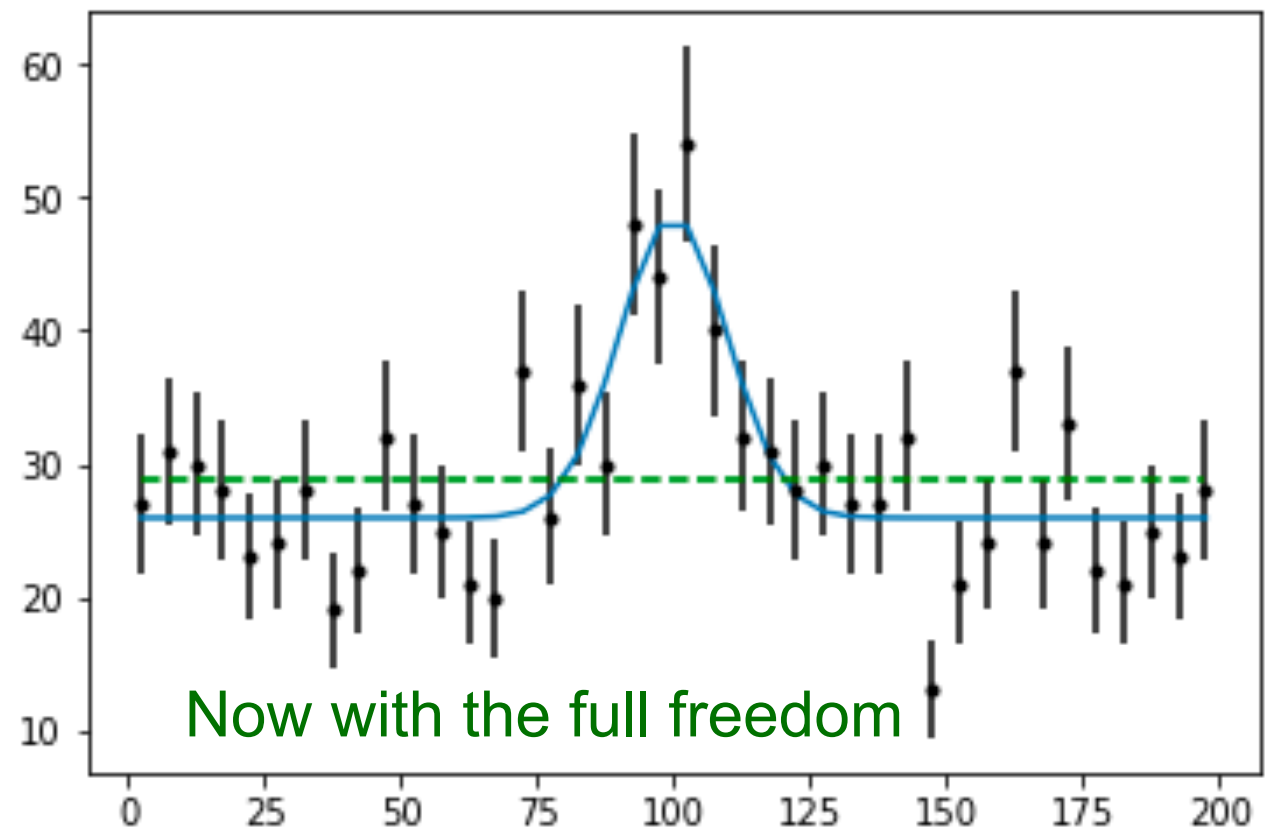
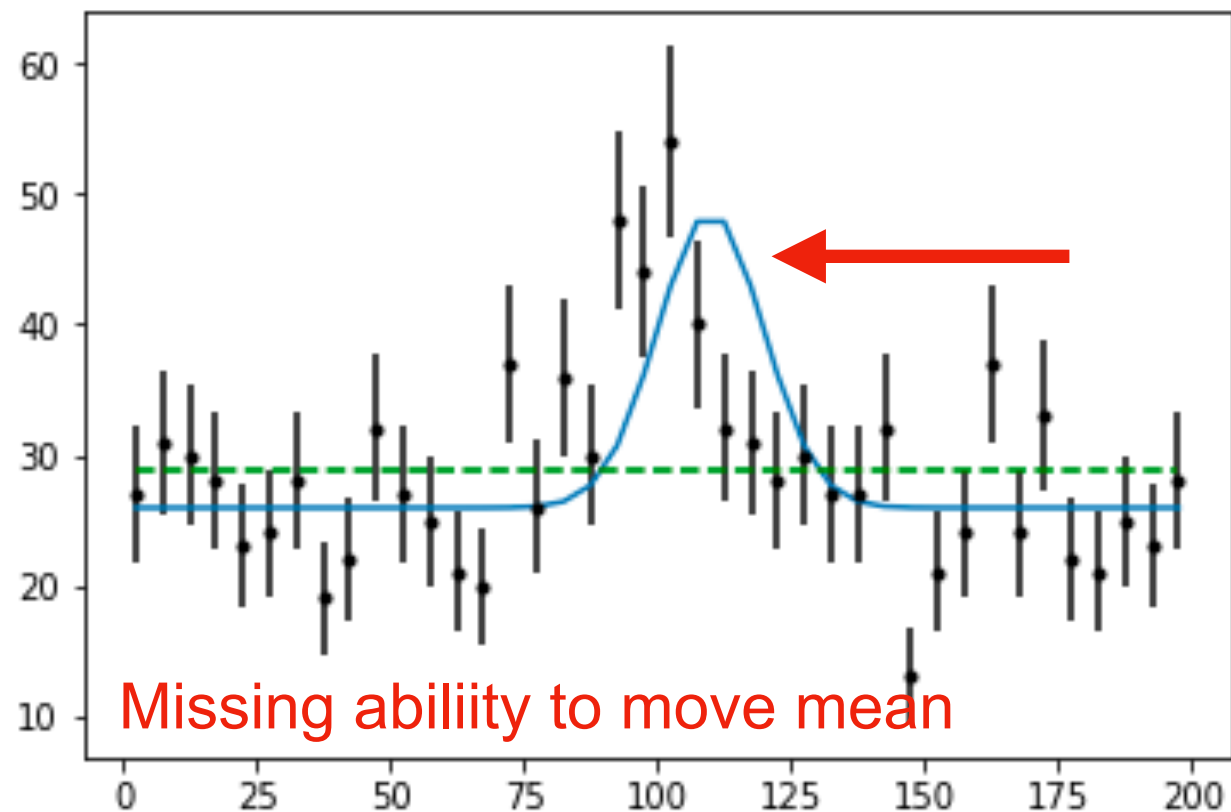




Lecture 10: Hypothesis Testing

What if models is² incomplete?

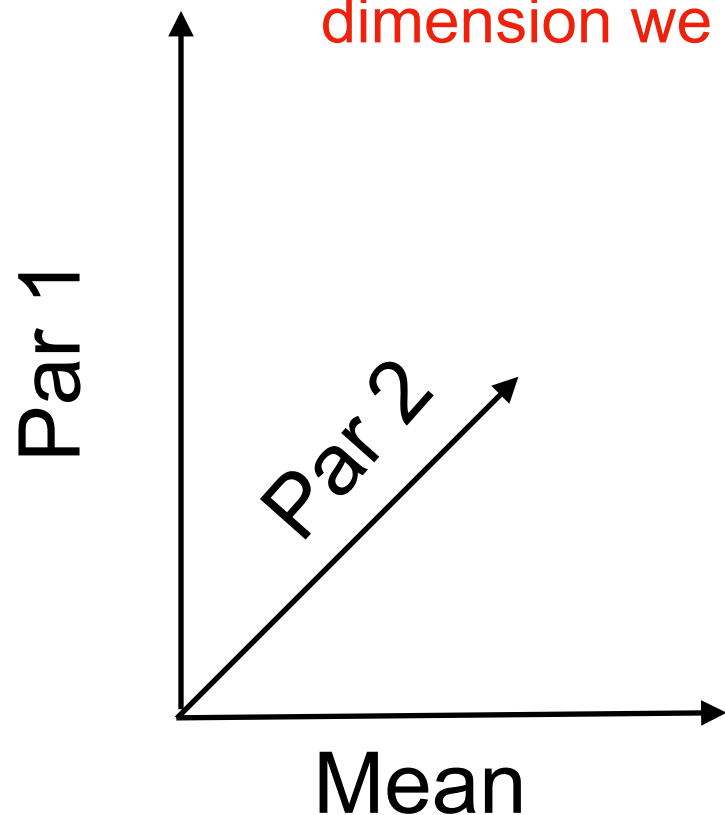
- <https://www.youtube.com/watch?v=Ku42Iszh9KM>
- Sometimes there are other parameters not included in our model



What if models is³ incomplete?

- <https://www.youtube.com/watch?v=Ku42lszh9KM>
- Sometimes there are other parameters not included in our model

Bayesian view:
Each parameter adds a
dimension we probe

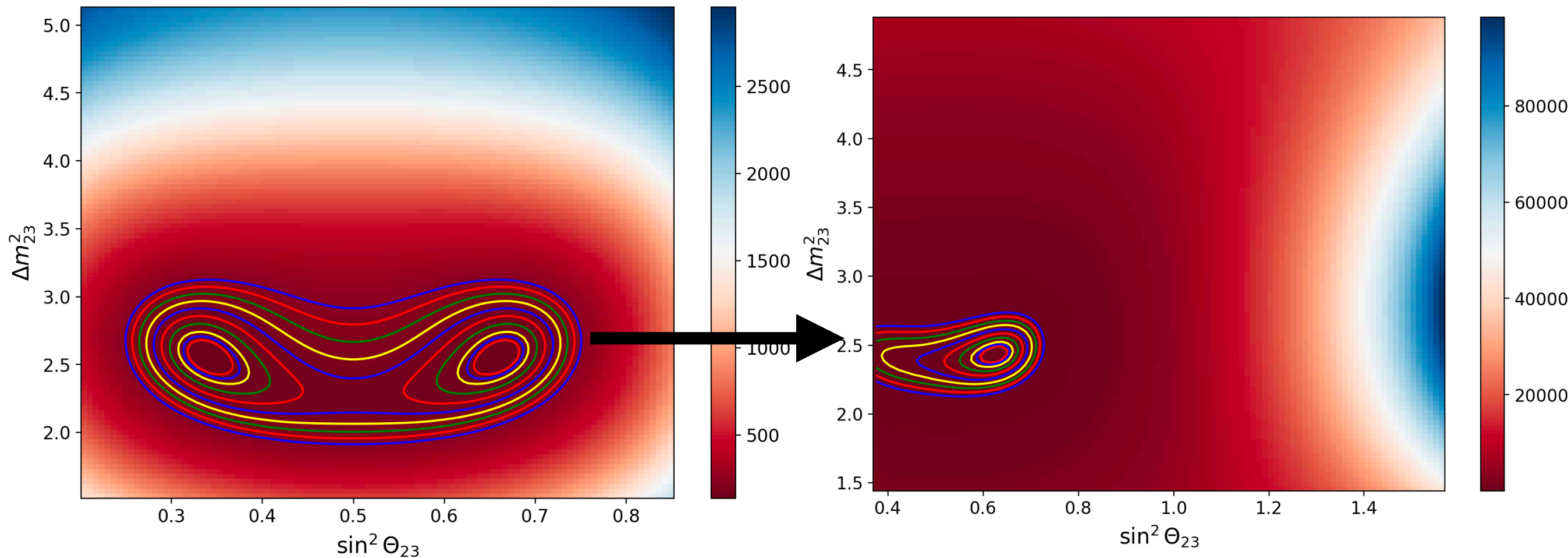


Frequentist View:
Make sure your model is
complete

$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right).$$

Float this parameter

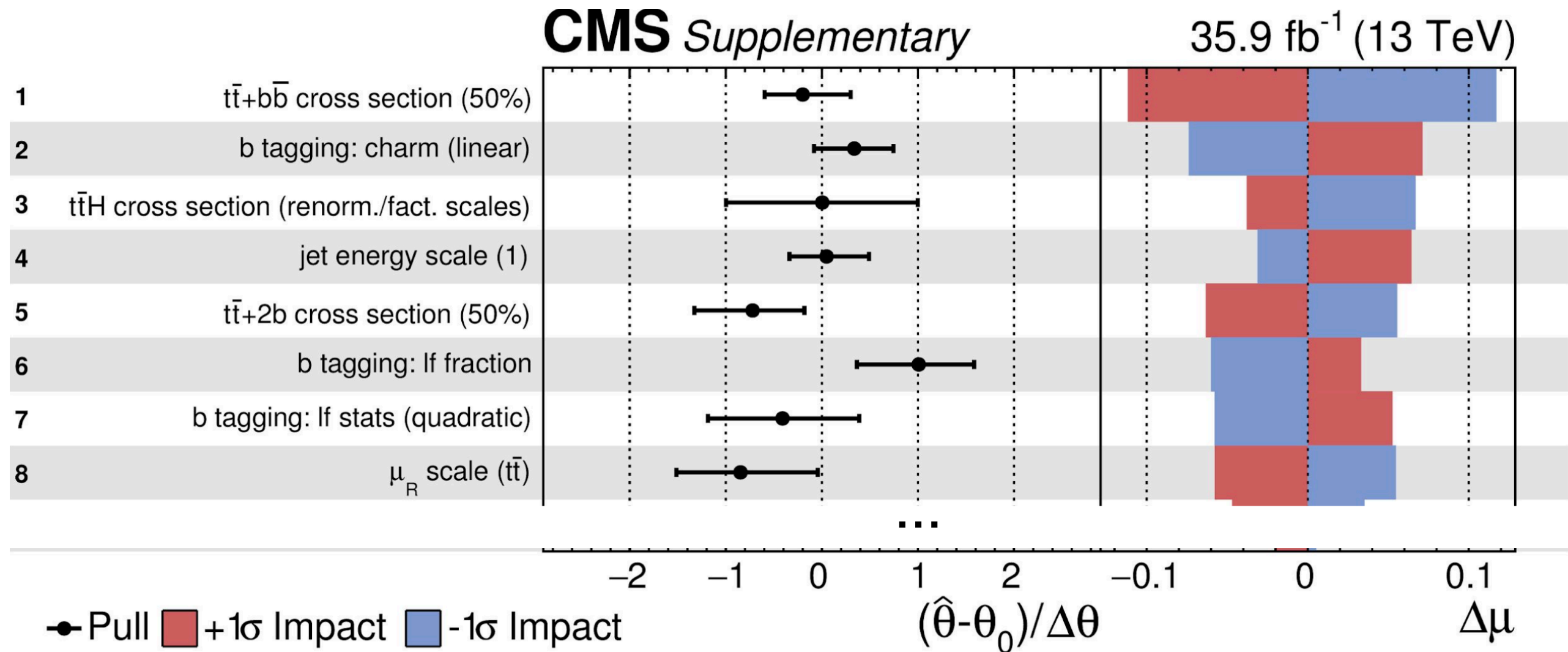
Nuisance



External Nuisances

$$-2 \log \left(\mathcal{L}(x|\vec{\theta}) \right) = -2 \log \left(\mathcal{L}(x|\vec{\theta}) \right)_{\text{original}} + \frac{(\sin \theta_{23} - \sin \theta_{23}^{\text{best}})^2}{\sigma_{\sin \theta_{23}}^2} + \frac{(\Delta m_{23}^2 - \Delta m_{23}^{2 \text{ best}})^2}{\sigma_{\Delta m_{12}^2}^2}$$

Nuisance



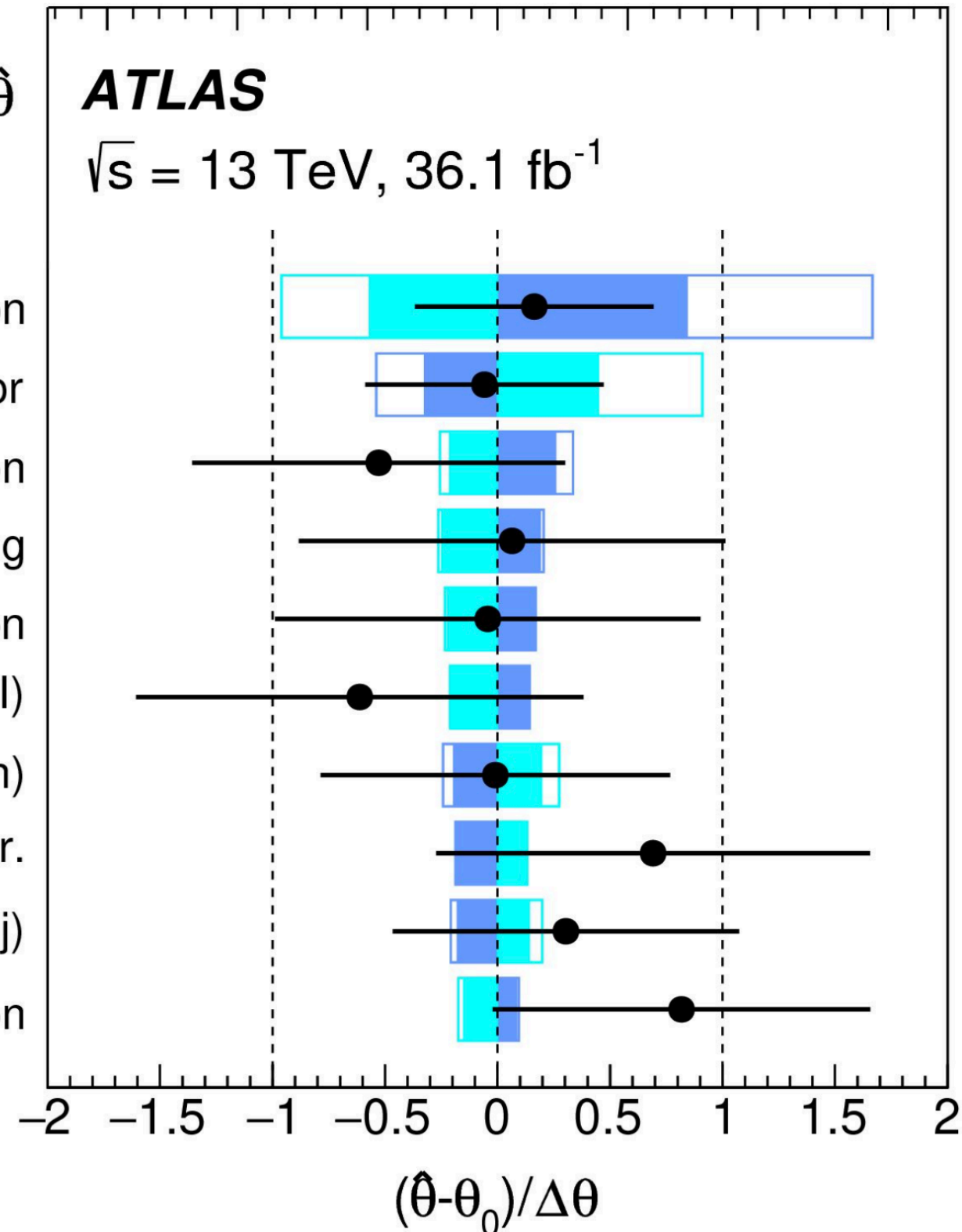
Nuisance

Post-fit impact on μ :

■ $\theta = \hat{\theta} + \Delta\hat{\theta}$ ■ $\theta = \hat{\theta} - \Delta\hat{\theta}$

—●— Nuis. Param. Pull

$t\bar{t}$ +jets PS and hadronization
 $t\bar{t}$ +jets NLO generator
 $t\bar{t}+\geq 1c$ normalization
 $t\bar{t}+\geq 1b$ NLO reweighting
 $t\bar{t}+\geq 1b$ MPI normalization
 b -tagging efficiency (NP I)
 JES (flavor composition)
 $t\bar{t}$ +light $/\geq 1c$ NNLO top- p_T corr.
 W/Z +jets normalization (9j)
 $t\bar{t}+\geq 1b$ normalization



Hypothesis Test

- Null Hypothesis :

Prior Posterior

$p_a(x)$ $p_b(x)$

- $p_a(x) = p_b(x)$

- Alternative Hypothesis

- $p_a(x) \neq p_b(x)$

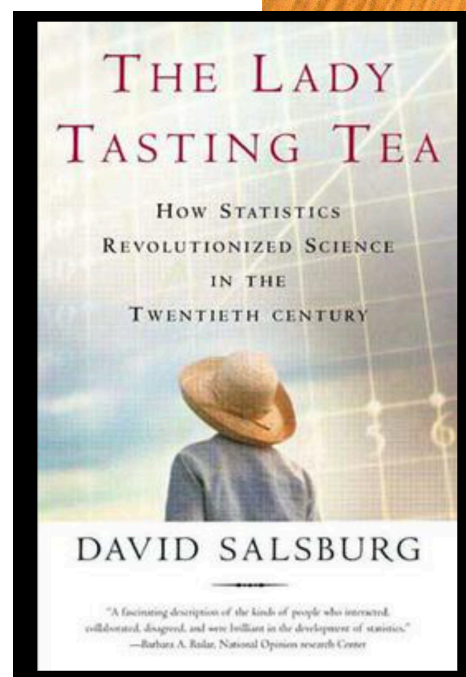
- Answers these questions:

- **Is my data consistent?**

- **Can I detect an effect?**

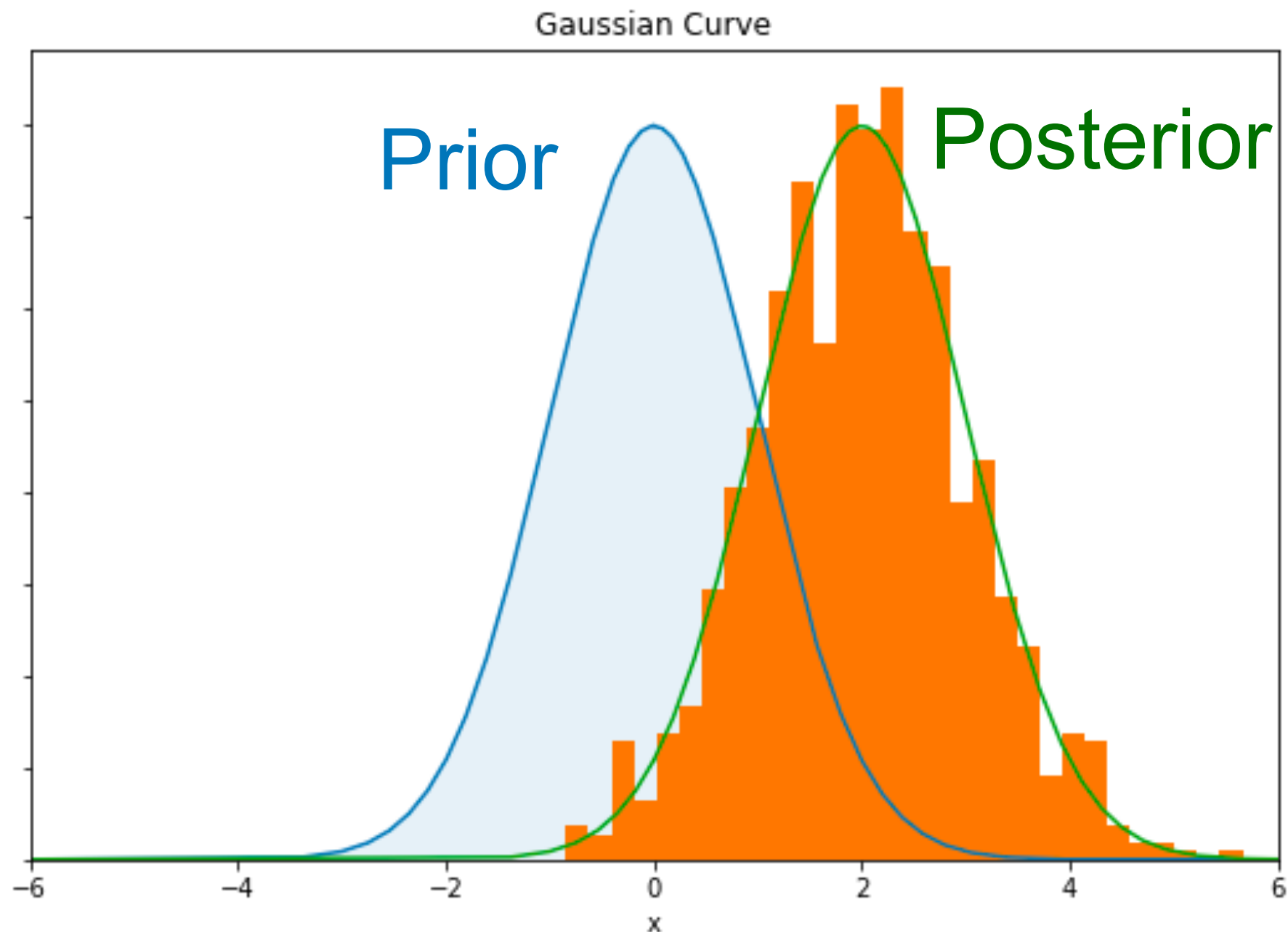
- **Did I just make a discovery?**

Hypothesis Test



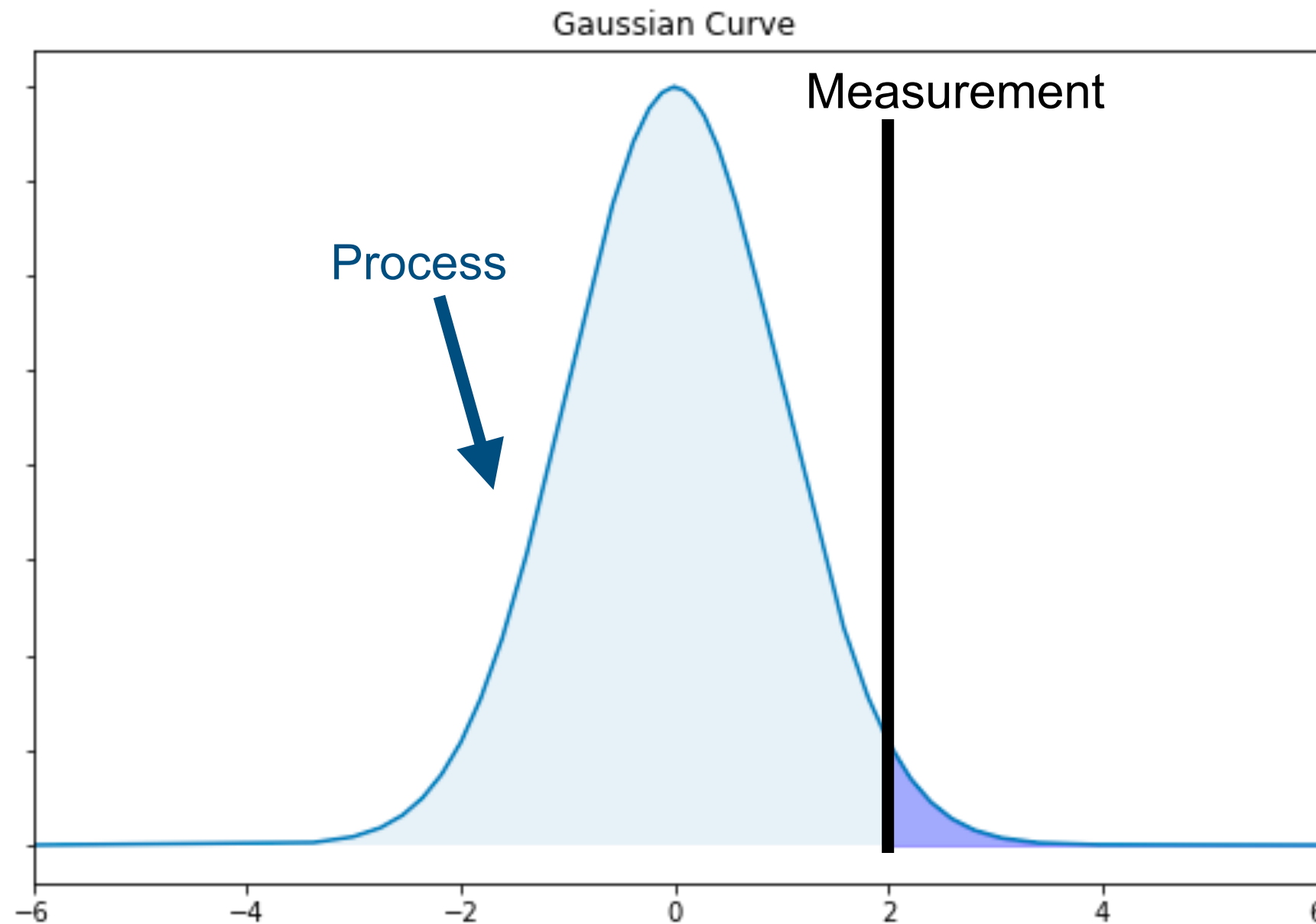
Lets Get Back to the⁹ original

- How do I tell if my measurements are consistent?



What is a measurement?

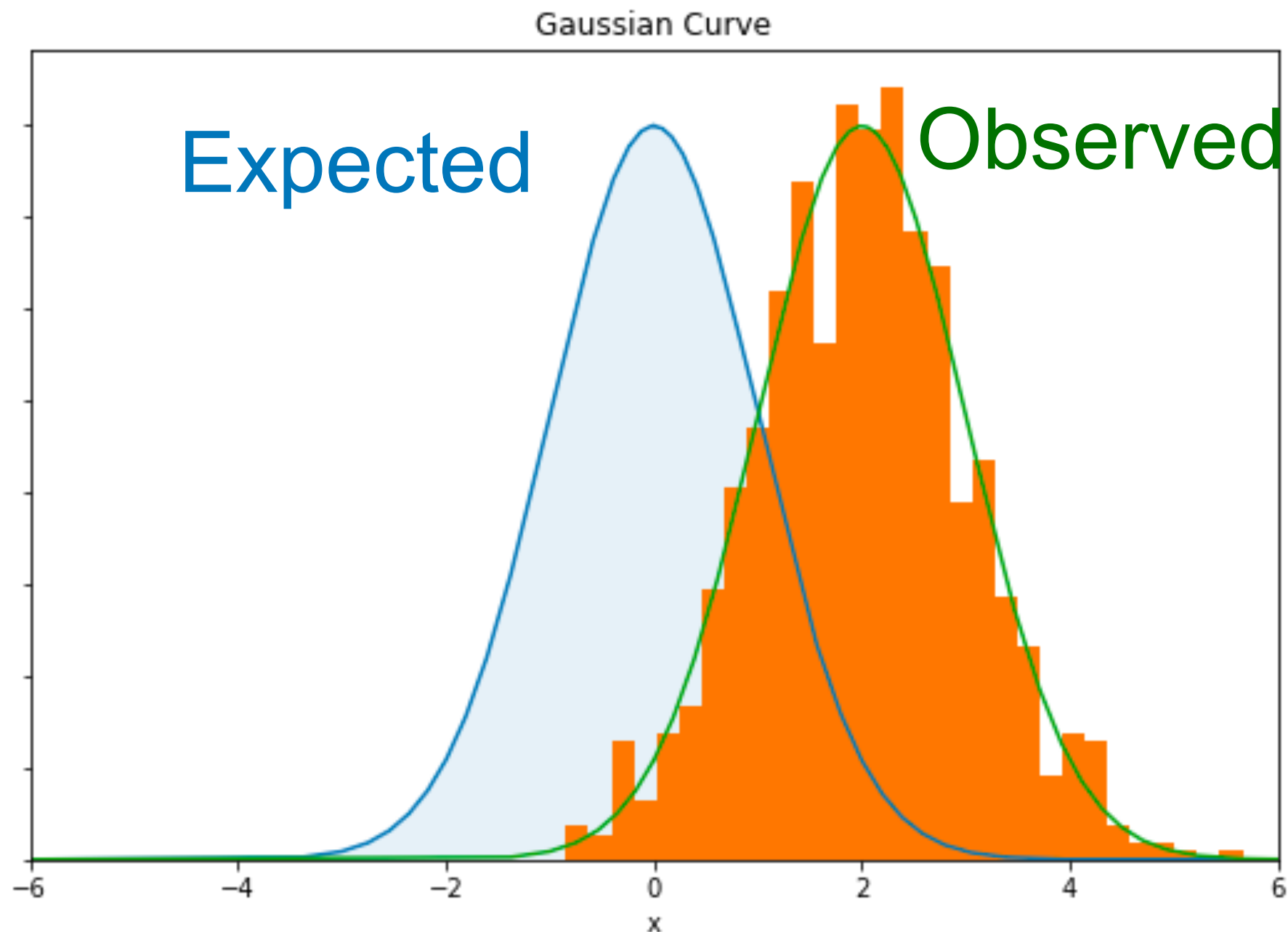
- We have assumed a model for this process
- With one measurement we can test it against a prior



Given our Prior we say that this measurement has a z-score of 2

Think Like a Frequentist

- How do I tell if my measurements are consistent?



Frequentist: chi2-test

- To test if two histograms are consistent

- $$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = N \sum_{i=1}^n \frac{(O_i/N - p_i)^2}{p_i}$$

where

χ^2 = Pearson's cumulative test statistic, which asymptotically approaches a χ^2 [distribution](#).

O_i = the number of observations of type i .

N = total number of observations

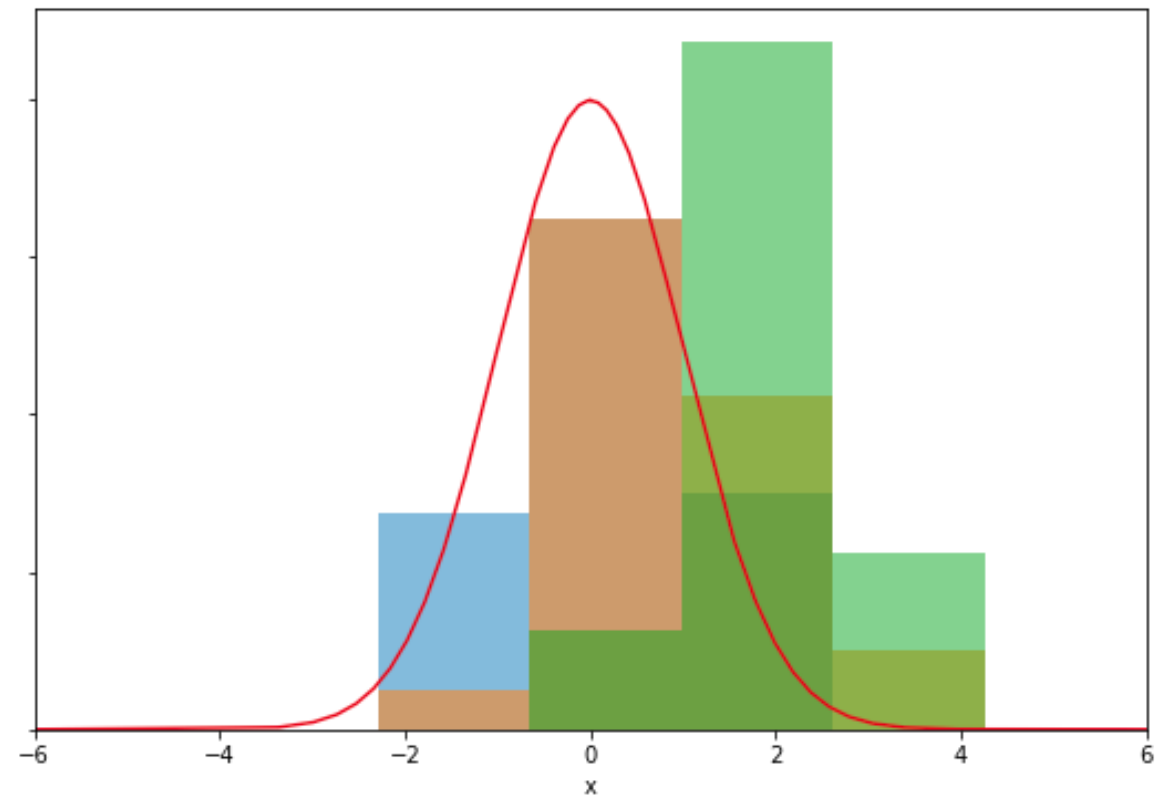
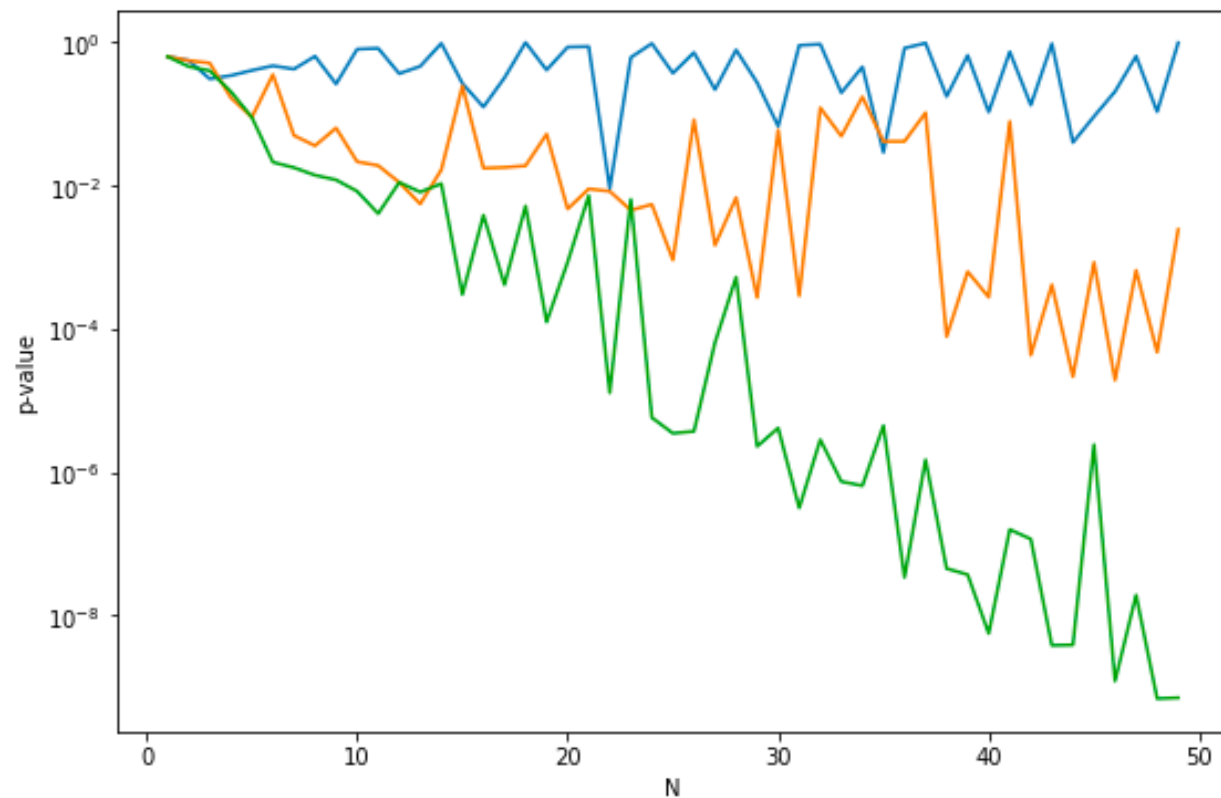
$E_i = Np_i$ = the expected (theoretical) count of type i , asserted by the null hypothesis that the fraction of type i in the population is p_i

n = the number of cells in the table.

If we have a continuous distribution we can turn it into a histogram with zero uncertainty, there are a lot of ways to do this

chi2 consistency

- Variation as a function of sample size
 - Compute chi2 consistency between histogram and observed
 - Larger p-value with larger n-samples



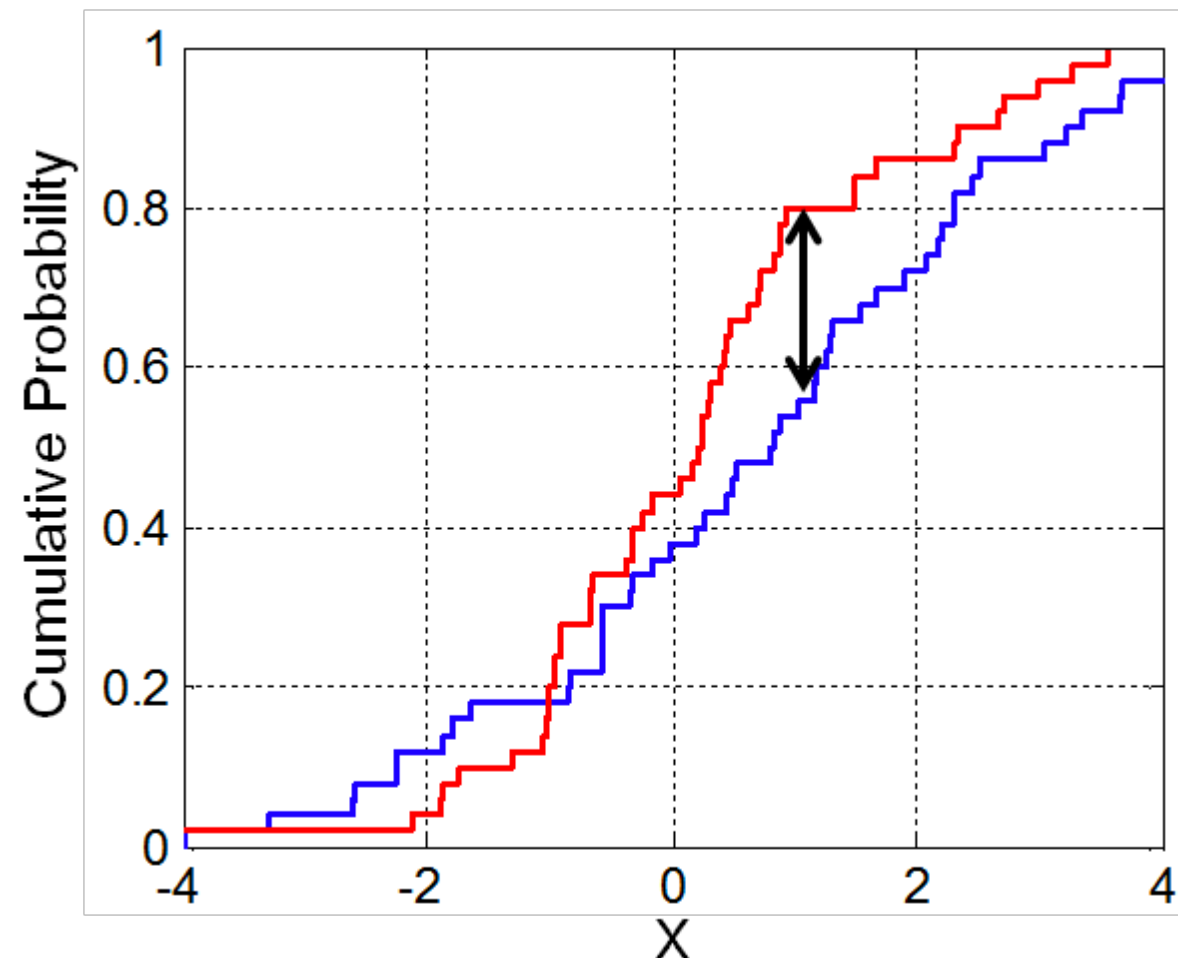
KS-test

- Komolgorov-Smirnov Test
- Estimate the maximum difference between observed and predicted cumulative distribution functions and compare with expectations.
- An unbinned way to check histograms are equal

$$\hat{F}_n(t) = \frac{\text{number of elements in the sample} \leq t}{n} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{X_i \leq t},$$

$$D_{n,m} = \sup_x |F_{1,n}(x) - F_{2,m}(x)|,$$

Supremum Function (ie max difference)



Likelihood Ratio Test

- Perhaps the most important test is the likelihood ratio test
- This is what we will focus for the rest of these slides
 - Likelihood ratio is the standard in hypothesis testing
 - This approach is very similar to a chi2 test

$\mathcal{L}(\theta_i) \forall \theta_i \in \Theta_0$ where Θ_0 refers to null – hypothesis

$\mathcal{L}(\theta_i) \forall \theta_i \in \Theta$ where Θ refers to alternative – hypothesis

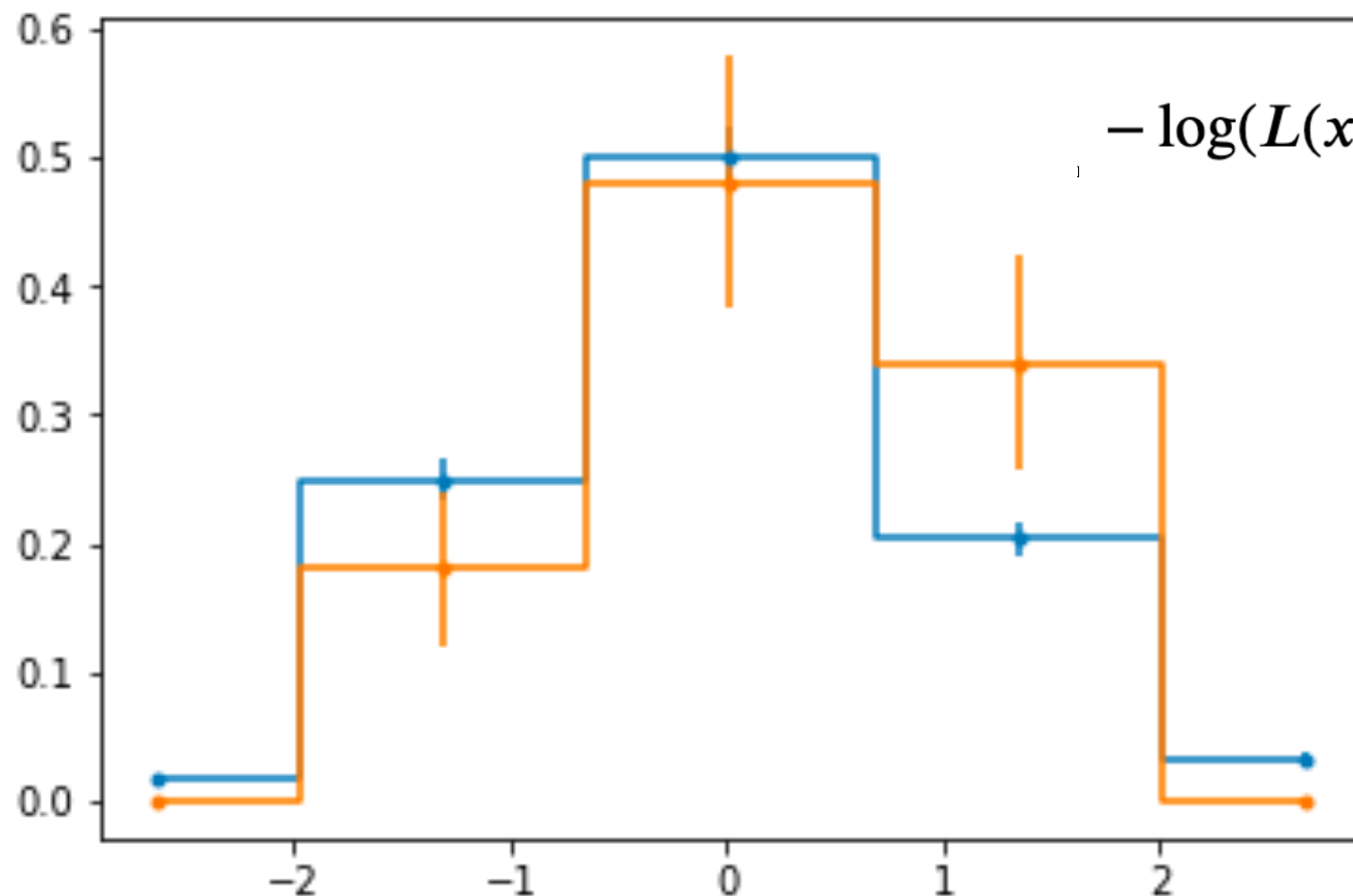
$\sup_{\theta \in \Theta_0} \mathcal{L}(\theta)$ is the maximum likelihood over all θ

$$\lambda_{\text{LR}} = -2 \ln \left[\frac{\sup_{\theta \in \Theta_0} \mathcal{L}(\theta)}{\sup_{\theta \in \Theta} \mathcal{L}(\theta)} \right]$$

Neyman-Pearson
Lemma states the
likelihood ratio
converges to the
optimal test statistic

Likelihood Test: Concept

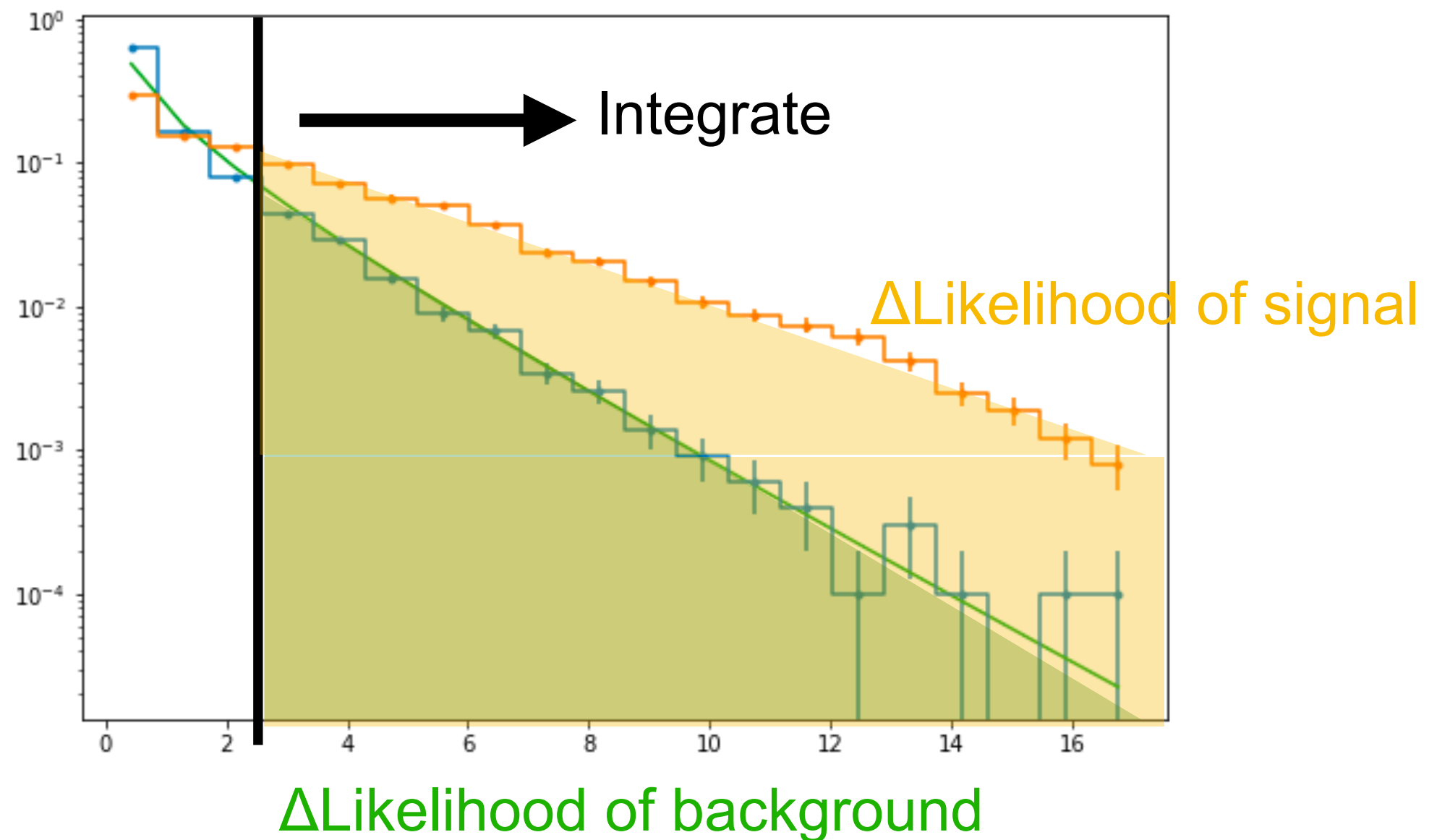
- We would like to see if these two histograms are consistent
 - Hypothesis: Orange is a gaussian with mean 0 (blue)
 - True: Orange is sampled from a gaussian with mean 0.2



$$-\log(L(x|\lambda)) = \sum_{i=1}^N -\frac{1}{2}\log(2\pi\sigma^2) + \frac{(x_i - \mu)^2}{2\sigma^2}$$

Likelihood Test: Gaussian

- We can see how these on average a deviation



T-test



Sample A



Sample B



Sample C

Are Sample means A, B, and C consistent with each other?

Frequentist: t-test

- Perform a t-test:
 - If samples $X_1, X_2, \dots, X_N$ from the same underlying distribution
 - Underlying distribution is Gaussian with mean μ and variance σ^2
 - If the above is true then we have a measured mean+variance:

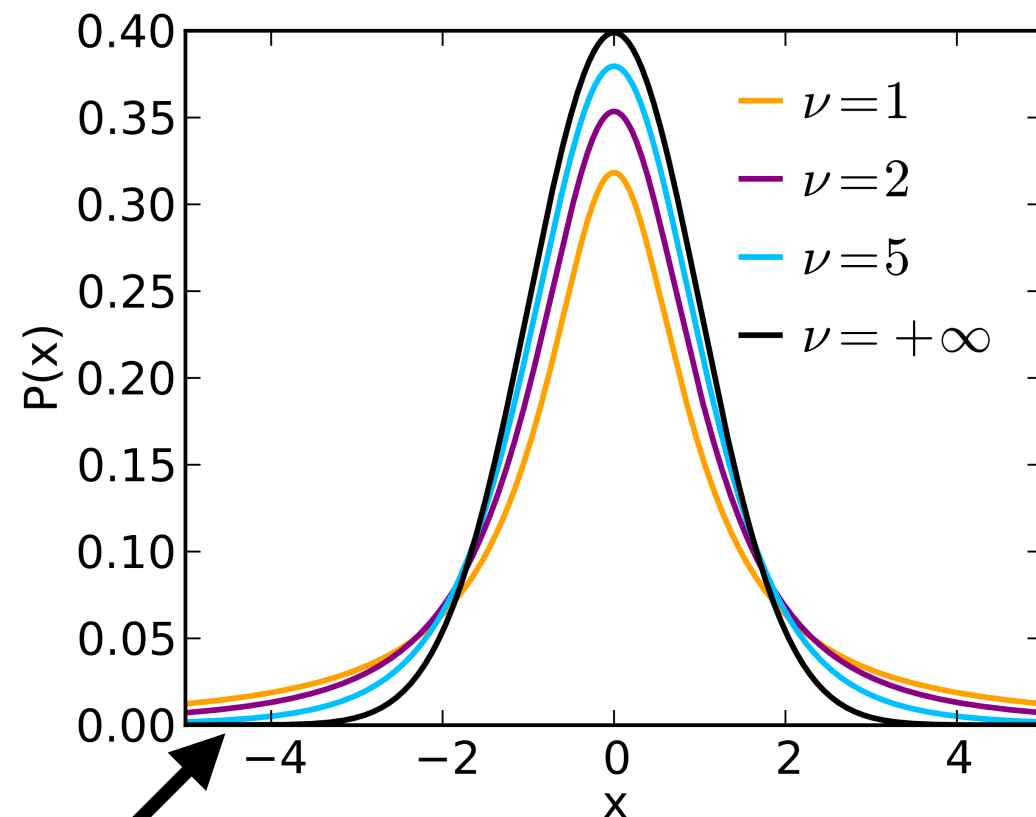
- given by $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$ and $S^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2$

- And the distribution $\frac{\bar{X} - \mu}{S/\sqrt{N}}$ is given by a t-distribution

Frequentist :t-test

- Often called the “students” t-test (<https://www.youtube.com/watch?v=U9Wr7VEPGXA>)
- Invented by brewmaster to test flavor variations in Guinness

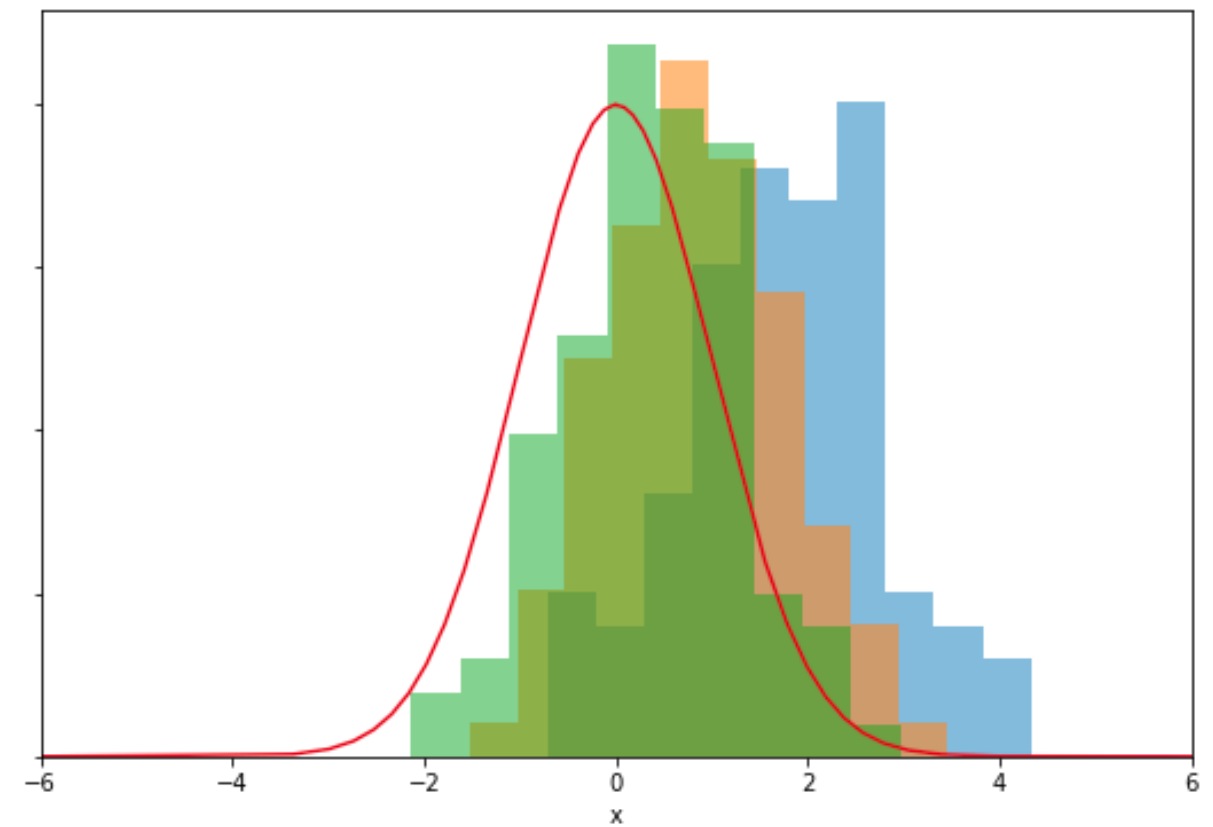
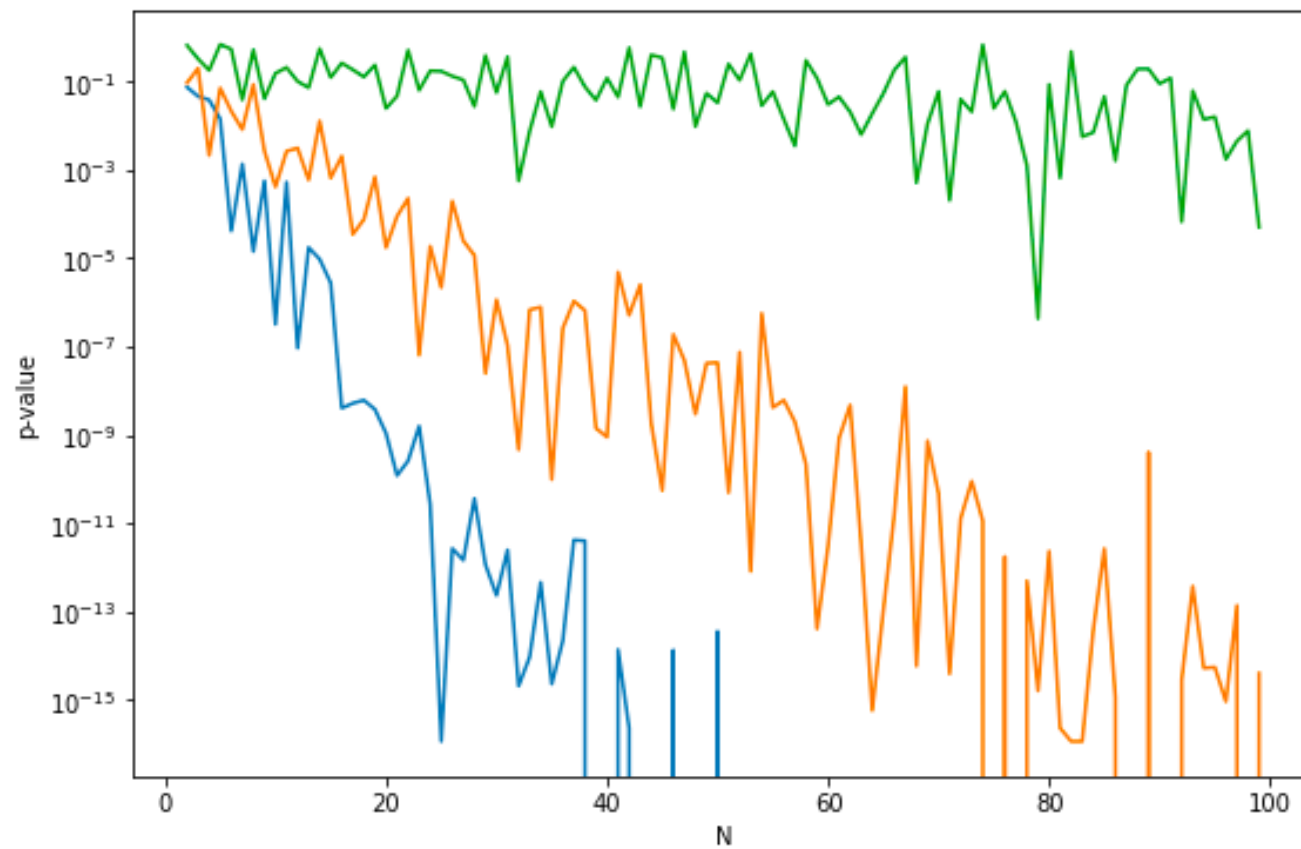
$$f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad \nu = N - 1$$



Normal Distribution

t-test Consistency

- Variation as a number of events



t-test Consistency²²

